LinkedIn edunet foundation

AI CAREERS FOR WOMEN

LINKEDIN FELLOWSHIP

AI Powered Sales Business Intelligence Dashboard

Project Domain

Excel & Power BI with Copilot

A Project Report

submitted in partial fulfilment of the requirements of the AICW Project

by:

Student Name	Student Email	Guide
Dhanalakshmi Chelladurai	**Dhanalakshmichelladurai8@gmail.com**	**Umamaheswari**

Institution:
Smt. Maniben M.P. Shah Women's College Of Arts & Commerce

Year - 2026

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to everyone who supported me during the completion of this project.

First, I would like to thank my project guide, **Mrs. Umamaheswari,** for her valuable guidance, encouragement, and continuous support throughout the project. Her suggestions and feedback helped me understand the topic better and complete the project successfully.

I would also like to thank **Edunet Foundation, Microsoft, LinkedIn, and SAP** for providing this fellowship opportunity and the learning resources that helped me gain knowledge about Artificial Intelligence and data analytics tools.

I am grateful to Smt. Maniben M.P. Shah Women's College Of Arts & Commerce for providing the facilities and support required to carry out this project.

Finally, I would like to thank my family and friends for their motivation, encouragement, and support during the entire project work.

ABSTRACT OF THE PROJECT

This project is part of the AICW Fellowship Program, a collaborative initiative by Edunet Foundation in partnership with Microsoft, LinkedIn, and SAP. The fellowship focuses on building practical skills in Artificial Intelligence, Data Analytics, and Business Intelligence.

This project focuses on analyzing a Retail Sales dataset to generate meaningful business insights. The main problem identified was that the raw dataset contained inconsistencies such as duplicate records, missing values, and unstructured data, which made analysis difficult. The objective of this project was to clean and prepare the dataset, perform analysis, and present insights through interactive dashboards and an AI-based solution.

Microsoft Excel was used for data cleaning tasks such as removing duplicates, handling missing values, and organizing the dataset. Power BI was used to build interactive dashboards that highlight key metrics such as sales trends, product performance, and regional analysis. In addition, Python was used to develop a simple chatbot that allows users to interact with the data and gain insights more efficiently.

The project resulted in clear visualization of sales performance and improved understanding of business patterns. The integration of a chatbot enhanced user interaction with the data. Overall, this project demonstrates how data cleaning, visualization, and basic AI tools can be combined to support better decision-making.

TABLE OF CONTENTS

Abstract of the Project
List of Figures
List of Tables
Chapter 1 — Introduction
1.1 Problem Statement
1.2 Motivation
1.3 Objectives
1.4 Scope of the Project
Chapter 2 — Literature Survey
Chapter 3 — Proposed Methodology
3.1 System Design
3.2 Modules / Components Used
3.3 Data Flow Diagram (DFD)
3.4 Advantages
3.5 Requirement Specification
Chapter 4 — Implementation and Results
4.1 Data Cleaning Results
4.2 Visualization / Output Results
4.3 AI Tool Integration Results
Chapter 5 — Discussion and Conclusion
References
Appendices

LIST OF FIGURES

Figure No.	Figure Title / Description	Page No.
Figure 1	*[Raw / uncleaned dataset screenshot]*	14
Figure 2	*[Cleaned dataset or validation summary screenshot]*	15
Figure 3	*[Main dashboard / report / output screenshot]*	16
Figure 4	*[Additional chart or visualization screenshot]*	16
Figure 5	*[AI tool (Copilot / Chatbot) in use screenshot]*	17
Figure 6	*[DFD Level 0 — Context Diagram]*	11
Figure 7	*[DFD Level 1 — Detailed Process Flow]*	11

LIST OF TABLES

Table No.	Table Title / Description	Page No.
Table 1	*[Raw dataset summary — field names, data types, row count]*	14
Table 2	*[Cleaned dataset summary — issues found and fixed]*	14
Table 3	*[Hardware Requirements]*	12
Table 4	*[Software Requirements]*	12
Table 5	*[Key findings / results summary]*	18

CHAPTER 1

Introduction

This chapter introduces the project undertaken as part of the AICW Fellowship Program. It outlines the background, the problem being addressed, the motivation behind this work, and the objectives and scope of the project. The fellowship projects span multiple domains including Data Analytics, Business Intelligence, Healthcare Analytics, Content Generation, and AI-assisted Reporting.

1.1 Problem Statement

Organizations in the retail sector generate large volumes of sales data on a daily basis. However, this data is often stored in raw form and contains issues such as duplicate records, missing values, and inconsistent formatting. This creates a major challenge in analyzing the data effectively and extracting meaningful insights.

This problem is significant because inaccurate or unclean data can lead to incorrect analysis, which directly affects business decisions, sales strategies, and overall profitability. Without proper data processing, organizations may fail to identify important trends, customer preferences, and high-performing products.

The primary users affected by this problem include business analysts, managers, and decision-makers who rely on accurate data to monitor performance and plan strategies. Retail companies, in particular, depend heavily on data-driven insights to stay competitive in the market.

Existing approaches often involve manual data handling or basic tools that are time-consuming and prone to errors. Additionally, many systems lack interactive visualization and user-friendly interfaces for exploring data. There is also limited use of AI-based tools, such as chatbots, which can simplify data interaction and improve accessibility for non-technical users.

1.2 Motivation

The motivation behind choosing this project comes from the increasing importance of data analysis in today's business environment and my interest in learning practical skills in this field. As a student of accounts, finance, and insurance, gaining knowledge about how data can be analyzed and used for decision-making is highly valuable for my future career.

This project is professionally meaningful as it allowed me to work with a real-world sample dataset obtained from Kaggle, which simulates actual retail sales data. It provided hands-on experience in data cleaning, analysis, and visualization. Additionally, developing a chatbot using Python helped me understand how AI can be integrated with data analytics to enhance user interaction.

The project has practical applications in the retail sector, where businesses rely on sales data to track performance, identify trends, and improve strategies. Even though the dataset is a sample, the techniques used in this project can be applied to real business data.

The impact of this project lies in demonstrating how raw data can be transformed into meaningful insights through proper cleaning and visualization. It also shows how AI tools like chatbots can make data more accessible and user-friendly.

This project aligns with modern industry trends where organizations are increasingly using data analytics, business intelligence tools like Power BI, and AI-based solutions to make informed decisions and gain a competitive advantage.

1.3 Objectives

1. To collect and understand the structure of the retail sales dataset obtained from Kaggle.

2. To clean and preprocess raw data using Microsoft Excel by removing duplicates, handling missing values, and correcting data formats.

3. To perform data analysis using Excel and Power BI to identify key patterns, trends, and insights in the dataset.

4. To build interactive dashboards in Power BI that effectively present sales performance, product analysis, and regional insights.

5. To develop a simple AI-based chatbot using Python to enable user interaction with the data and provide quick insights.

6. To enhance decision-making by transforming raw data into meaningful and actionable insights through visualization and AI tools.

1.4 Scope of the Project

This project covers data from a retail sales dataset obtained from Kaggle, which includes information related to sales transactions, products, and regional performance.

The tools used in this project are limited to Microsoft Excel for data cleaning and preprocessing, Power BI for data visualization and dashboard creation, and Python for developing a basic AI chatbot.

The analysis mainly focuses on data cleaning, data transformation, visualization, and generating insights through interactive dashboards. It also includes basic AI integration through a chatbot to improve user interaction with the data.

The project does not include real-time data integration, advanced predictive analytics, or machine learning models. It is limited to descriptive analysis and basic AI functionality.

This project is intended as part of a fellowship showcase and is not designed as a production-level system.

CHAPTER 2

Literature Survey

A literature survey reviews existing research, tools, and methodologies relevant to the project domain. It helps identify gaps that the current project aims to address.

2.1 Existing Work and Research

In the domain of data analytics and business intelligence, several tools and techniques are widely used for handling and analyzing retail sales data. Microsoft Excel is commonly used for data cleaning and preprocessing, where techniques such as filtering, sorting, removal of duplicates, and handling missing values help improve data quality. According to Microsoft Excel documentation, these methods are effective for basic data preparation tasks.

Power BI is another widely used tool for data visualization and dashboard creation. As per Microsoft Power BI documentation, interactive dashboards help organizations monitor key performance indicators and make informed decisions. Many businesses use pre-built templates and visual reports to analyze sales trends and performance.

In recent years, Artificial Intelligence has been increasingly integrated into data analysis. Research on AI-assisted tools and prompt-based systems shows that chatbots and virtual assistants can improve accessibility and simplify data interaction, especially for non-technical users. Studies on AI prompt engineering highlight how structured queries can generate better and more relevant outputs.

However, existing solutions have certain limitations. Excel-based data cleaning can be time-consuming and less efficient for large datasets. Power BI dashboards, while powerful, may require technical knowledge to interpret effectively. Additionally, many existing systems lack integration with AI-based tools, limiting user interaction and accessibility.

This project addresses these gaps by combining multiple approaches into a single workflow. It integrates Excel for efficient data cleaning, Power BI for interactive visualization, and a Python-based chatbot to enhance user interaction. This combination improves usability, reduces complexity, and demonstrates how AI can be effectively integrated with traditional data analysis tools.

2.2 Tools and Technologies Review

The following tools and technologies were used in this project for data cleaning, analysis, visualization, and AI integration:

- **Microsoft Excel** — Used for data cleaning and preprocessing. Techniques such as removing duplicates, handling missing values, sorting, filtering, and basic data transformation were applied to prepare the dataset for analysis.

- **Microsoft Power BI** — A business intelligence tool used to create interactive dashboards and visualizations. It helped in analyzing sales trends, product performance, and regional insights through user-friendly reports.

- **Python** — Used to develop a basic AI chatbot. Python programming enabled the creation of a simple system where users can interact with the data and obtain insights in an easy and efficient manner.

- **AI Chatbot** — Implemented using Python to enhance user interaction. It allows users to ask questions related to the dataset and receive relevant responses, making data analysis more accessible, especially for non-technical users.

- **Kaggle Dataset** — A publicly available retail sales dataset was used for this project. It provides sample real-world data for practicing data analysis, cleaning, and visualization techniques.

2.3 Gaps Identified

- Most existing research focuses on advanced machine learning and predictive models, while simpler tool-based solutions using Excel and Power BI for non-technical users are less explored.

- There is limited practical implementation of integrating AI-based chatbots with data analysis tools like Excel and Power BI in student-level projects.

- Many existing dashboards provide visual insights but lack interactive features that allow users to directly query the data in a simple and user-friendly manner.

- Data cleaning processes in tools like Excel are often manual and time-consuming, especially when dealing with large datasets, and lack automation support.

- There is a lack of unified approaches that combine data cleaning, visualization, and AI-based interaction in a single workflow for better usability and accessibility.

CHAPTER 3

Proposed Methodology

This chapter describes the methodology adopted to achieve the project objectives. It covers the system design, modules used, data flow, advantages, and requirement specifications.

3.1 System Design

The system design of this project follows a structured approach where raw data is processed step-by-step to generate meaningful insights. The flow of the system moves from data input to cleaning, analysis, and final output through dashboards and an AI chatbot.

• **Data Source / Input:**
The project uses a retail sales dataset obtained from Kaggle in CSV/Excel format. The dataset contains sales transaction details such as product information, sales amount, and regional data, which serves as the raw input for analysis.

• **Data Cleaning:**
Data cleaning was performed using Microsoft Excel. This included removing duplicate records, handling missing values, correcting inconsistent data formats, and organizing the dataset into a structured form suitable for analysis.

• **Analysis / Processing:**
The cleaned data was analyzed using Excel and Power BI. Techniques such as sorting, filtering, pivot tables, and basic calculations were used in Excel. In Power BI, data modeling and measures were applied to identify trends, patterns, and key performance indicators.

• **Visualization / Output:**
The final output of the project includes interactive dashboards created in Power BI, which display insights such as sales trends, product performance, and regional analysis. Additionally, a Python-based AI chatbot was developed to allow users to interact with the data and obtain insights in a simple and user-friendly manner.

3.2 Modules / Components Used

• **Module 1 — Data Ingestion:**
Loading the retail sales dataset obtained from Kaggle into Microsoft Excel and Power BI for further processing and analysis.

• **Module 2 — Data Cleaning:**
Performing data preprocessing in Microsoft Excel by removing duplicate records, handling missing values, correcting data formats, and organizing the dataset into a structured form.

• **Module 3 — Data Analysis:**
Analyzing the cleaned data using Excel (sorting, filtering, pivot tables) and Power BI (data modeling and DAX measures) to identify key patterns, trends, and performance indicators.

• **Module 4 — Visualization / Report Generation:**
Creating interactive dashboards in Power BI to present insights such as sales trends, product performance, and regional analysis in a clear and user-friendly format.

• **Module 5 — AI Integration:**
Developing a Python-based chatbot that allows users to interact with the dataset and obtain insights through simple queries, improving accessibility and ease of use.

• **Module 6 — Insight Generation:**
Summarizing key findings from the analysis to support better decision-making and highlight important business insights derived from the dataset.

3.3 Data Flow Diagram (DFD)

DFD Level 0 — Context Diagram

Input	Process	Output
Retail Sales Dataset	Retail Sales Analysis System	Power BI Dashboards + AI Chatbot Output

DFD Level 1 — Detailed Process Flow

Process Step	Descriptions
Data Input	Retail Sales Dataset Obtained from Kaggle
Data Loading	Dataset Loaded into Excel and Power BI
Data Cleaning	Removing Duplicates, handling missing values, correcting formats
Data Analysis	Using Excel and Power BI to identify trends and patterns
Visualization	Creating dashboards in Power BI
AI Chatbot	Python based Chatbot for user interaction
Output	Final insights and reports generated

3.4 Advantages

- Reduces manual effort by using Excel for efficient data cleaning tasks such as removing duplicates and handling missing values.

- Provides clear and interactive visual insights through Power BI dashboards, making it easy for non-technical users to understand business performance.
- Enhances user interaction by integrating a Python-based AI chatbot, allowing users to quickly access insights through simple queries.
- Improves decision-making by converting raw retail sales data into meaningful and actionable insights.
- Scalable — the same approach can be applied to larger datasets or different domains such as finance, healthcare, or marketing.
- Cost-effective — uses widely available tools like Microsoft Excel and Power BI, which are commonly used in industry.
- Demonstrates practical integration of data analytics and AI, making the project relevant to modern industry trends.

3.5 Requirement Specification

Hardware Requirements

Component	Specification
Processor	Intel Core i3
RAM	Minimum 4 GB
Storage	Minimum 50 GB free disk space
Display	1366 x 768 resolution or higher
Internet	Required for Chatbot access

Software Requirements

Software	Details
Operating System	Windows 11
Microsoft Excel	Excel 2019
Power BI Desktop	Latest version
Programming Language	Python
AI Chatbot	ChatGPT
Web Browser	Google Chrome & Microsoft Edge

CHAPTER 4

Implementation and Results

This chapter presents the actual implementation of the project and the results obtained at each stage. Screenshots of dashboards cleaned datasets, or generated content should be inserted where indicated.

4.1 Data Cleaning Results

The dataset used in this project was obtained from Kaggle and consists of retail sales transaction data.

The original dataset contained multiple rows and columns representing sales details such as products, regions, and sales values.

During initial analysis, several issues were identified in the dataset. These included duplicate records, missing values in certain fields, inconsistent data formats, and unstructured data, which made analysis difficult.

To address these issues, data cleaning was performed using Microsoft Excel. Duplicate entries were removed to avoid redundancy. Missing values were handled appropriately, and inconsistent data formats were corrected to maintain uniformity. The dataset was also organized and structured properly for further analysis.

After applying these cleaning steps, the dataset became more accurate, consistent, and suitable for analysis. The cleaned dataset contains a refined set of records with improved data quality.

Attribute	Description
Dataset Source	Kaggle
Number of Rows	9994
Number of Columns	21
Key Fields	Order ID, Customer Name, Product Name, Sales, Profit, Quantity, Region
Data Types	Numerical (Sales, Profit), Categorical (Region, Category), Text (Customer Name, Product Name)

1	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region	Product ID	Category	Sub-Category	Product Name	Sales	Quantity	Discount	Profit
2	1	CA-2016-152156	11-08-2016	11-11-2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42420	South	FUR-BO-10001798	Furniture	Bookcases	Bush Somerset Collection Bookcase	261.96	2	0	41.9136
3	2	CA-2016-152156	11-08-2016	11-11-2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42420	South	FUR-CH-10000454	Furniture	Chairs	Hon Deluxe Fabric Upholstered Stacking Chairs, Rounded Back	731.94	3	0	219.582
4	3	CA-2016-138688	06-12-2016	6/16/2016	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	California	90036	West	OFF-LA-10000240	Office Supplies	Labels	Self-Adhesive Address Labels for Typewriters by Universal	14.62	2	0	6.8714
5	4	US-2015-108966	10-11-2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311	South	FUR-TA-10000577	Furniture	Tables	Bretford CR4500 Series Slim Rectangular Table	957.5775	5	0.45	-383.031
6	5	US-2015-108966	10-11-2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311	South	OFF-ST-10000760	Office Supplies	Storage	Eldon Fold 'N Roll Cart System	22.368	2	0.2	2.5164
7	6	CA-2014-115812	06-09-2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	FUR-FU-10001487	Furniture	Furnishings	Eldon Expressions Wood and Plastic Desk Accessories, Cherry Wood	48.86	7	0	14.1694
8	7	CA-2014-115812	06-09-2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-AR-10002833	Office Supplies	Art	Newell 322	7.28	4	0	1.9656
9	8	CA-2014-115812	06-09-2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	TEC-PH-10002275	Technology	Phones	Mitel 5320 IP Phone VoIP phone	907.152	6	0.2	90.7152
10	9	CA-2014-115812	06-09-2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-BI-10003910	Office Supplies	Binders	DXL Angle-View Binders with Locking Rings by Samsill	18.504	3	0.2	5.7825
11	10	CA-2014-115812	06-09-2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-AP-10002892	Office Supplies	Appliances	Belkin F5C206VTEL 6 Outlet Surge	114.9	5	0	34.47
12	11	CA-2014-115812	06-09-2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	FUR-TA-10001539	Furniture	Tables	Chromcraft Rectangular Conference Tables	1706.184	9	0.2	85.3092
13	12	CA-2014-115812	06-09-2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	TEC-PH-10002033	Technology	Phones	Konftel 250 Conference phone - Charcoal black	911.424	4	0.2	68.3568
14	13	CA-2017-114412	4/15/2017	4/20/2017	Standard Class	AA-10480	Andrew Allen	Consumer	United States	Concord	North Carolina	28027	South	OFF-PA-10002365	Office Supplies	Paper	Xerox 1967	15.552	3	0.2	5.4432
15	14	CA-2016-161389	12-05-2016	12-10-2016	Standard Class	IM-15070	Irene Maddox	Consumer	United States	Seattle	Washington	98103	West	OFF-BI-10003656	Office Supplies	Binders	Fellowes PB200 Plastic Comb Binding Machine	407.976	3	0.2	132.5922
16	15	US-2015-118983	11/22/2015	11/26/2015	Standard Class	HP-14815	Harold Pawlan	Home Office	United States	Fort Worth	Texas	76106	Central	OFF-AP-10002311	Office Supplies	Appliances	olmes Replacement Filter for HEPA Air Cleaner, Very Large Room, HEPA Filt	68.81	5	0.8	-123.858

Figure 1: Raw (uncleaned) retail sales dataset

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
1	Category	City	Country	Customer ID	Customer Name	Sum of Discount	Order Date	Order ID	Count of Postal Code	Product ID	Product Name	Sum of Profit	Sum of Quantity	Region	Count of Row ID	Sum of Sales	Segment	Ship Mode	State	Sub-Category	Year	Quarter	Month	Day
2	Furniture	Akron	United States	EB-13705	Ed Braxton	0.4	20-10-2017 00:00	CA-2017-147277	1	FUR-TA-10001539	Chromcraft Rectangular Conference Tables	-75.8304	2	East	1	284.364	Corporate	Standard Class	Ohio	Tables	2017	Qtr 4	October	24
3	Furniture	Akron	United States	TT-21070	Ted Trevino	0.2	18-05-2014 00:00	CA-2014-164224	1	FUR-FU-10000508	Deflecto Glass Clear Studded Chair Mats	5.7508	5	East	1	148.232	Consumer	Second Class	Ohio	Furnishings	2014	Qtr 2	May	20
4	Furniture	Alexandria	United States	AG-10495	Andrew Gjertsen	0	04-01-2015 00:00	CA-2015-104241	1	FUR-FU-10001473	DAX Wood Document Frame	88.1992	14	South	1	192.22	Corporate	Standard Class	Virginia	Furnishings	2015	Qtr 1	January	9
5	Furniture	Alexandria	United States	SO-20485	Shirley Daniels	0	26-01-2014 00:00	US-2014-155502	1	FUR-FU-10001847	Eldon Image Series Black Desk Accessories	4.4712	3	South	1	12.42	Home Office	Standard Class	Virginia	Furnishings	2014	Qtr 1	January	31
6	Furniture	Alexandria	United States	SO-20485	Shirley Daniels	0	26-01-2014 00:00	US-2014-155502	1	FUR-FU-10004587	GE General Use Halogen Bulbs, 100 Watts, 1 Bulb per Pack	30.7818	3	South	1	62.82	Home Office	Standard Class	Virginia	Furnishings	2014	Qtr 1	January	31
7	Furniture	Allen	United States	AG-10675	Anna Gayman	0.3	07-05-2015 00:00	CA-2015-137512	1	FUR-TA-10001095	Chromcraft Round Conference Tables	-31.3722	2	Central	1	244.006	Consumer	Standard Class	Texas	Tables	2015	Qtr 2	May	12
8	Furniture	Allentown	United States	CJ-12010	Caroline Jumper	0.2	22-07-2016 00:00	CA-2016-122362	1	FUR-FU-10002456	Master Caster Door Stop, Large Neon Orange	3.3488	2	East	1	11.648	Consumer	Standard Class	Pennsylvania	Furnishings	2016	Qtr 3	July	27
9	Furniture	Amarillo	United States	DS-13180	David Smith	0.32	31-05-2017 00:00	CA-2017-154807	1	FUR-BO-10002814	Bush Mission Pointe Library	-56.2352	2	Central	1	205.3528	Corporate	Standard Class	Texas	Bookcases	2017	Qtr 2	April	4
10	Furniture	Amarillo	United States	JE-15745	Joel Eaton	0.3	15-10-2013 00:00	CA-2015-139731	1	FUR-CH-10002024	HON 5400 Series Task Chairs for Big and Tall	-550.49	5	Central	1	2455.43	Consumer	Same Day	Texas	Chairs	2015	Qtr 4	October	15
11	Furniture	Amarillo	United States	MM-18520	Naomi Murray	0.8	07-01-2016 00:00	CA-2016-158808	1	FUR-FU-10004270	Executive Impressions 13" Clairmont Wall Clock	-10.9611	3	Central	1	23.078	Consumer	Standard Class	Texas	Furnishings	2016	Qtr 1	January	11
12	Furniture	Amarillo	United States	NR-18550	Nick Radford	0.3	02-05-2016 00:00	CA-2016-109400	1	FUR-CH-10002298	Office Star - Contemporary Task Swivel chair with Loop Arms, Charcoal	-110.0232	4	Central	1	366.744	Consumer	Standard Class	Texas	Chairs	2016	Qtr 2	May	6
13	Furniture	Anaheim	United States	BP-11185	Ben Peterman	0	29-12-2017 00:00	CA-2017-146626	1	FUR-FU-10002501	Nu-Dell Executive Frame	37.4144	8	West	1	101.12	Corporate	Standard Class	California	Furnishings	2018	Qtr 1	January	5
14	Furniture	Anaheim	United States	EB-13705	Ed Braxton	0.1	14-06-2016 00:00	CA-2016-128223	1	FUR-TA-10001771	Bush Cubix Conference Tables, Fully Assembled	80.843	7	West	1	1293.488	Corporate	Same Day	California	Tables	2016	Qtr 2	June	14
15	Furniture	Anaheim	United States	KD-16615	Ken Dana	0.2	13-09-2015 00:00	US-2015-129007	1	FUR-CH-10000155	Global Comet Stacking Armless Chair	71.772	3	West	1	717.72	Corporate	First Class	California	Chairs	2015	Qtr 3	September	15
16	Furniture	Anaheim	United States	KD-16615	Ken Dana	0	13-09-2015 00:00	US-2015-129007	1	FUR-FU-10002379	Eldon Econoclear Chair Mats for Low Pile Carpets	24.882	5	West	1	207.35	Corporate	First Class	California	Furnishings	2015	Qtr 3	September	15

Figure 2: Cleaned dataset after preprocessing in Excel

4.2 Visualization / Output Results

The project produced multiple interactive dashboards using Power BI to present insights from the retail sales dataset. The dashboard is divided into different pages, each focusing on a specific aspect of analysis.

Dashboard / Page 1: Information
This page provides an overview of the dataset, including key details and summary metrics to give users a basic understanding of the data.

- **Dashboard / Page 2: Executive Overview**
  This page presents high-level insights such as total sales, total profit, and key performance indicators, helping decision-makers quickly understand overall business performance.

- **Dashboard / Page 3: Sales Analytics**
  This page shows detailed sales trends, including time-based analysis, product performance, and sales distribution, helping identify patterns and growth trends.
- **Dashboard / Page 4: Risks and Performance**
  This page highlights areas of low performance and potential risks, allowing businesses to identify weak points and take corrective actions.

- **Dashboard / Page 5: AI Insights**
  This page presents insights generated with the help of AI techniques, providing deeper understanding and quick summaries of important patterns in the data.

- **Dashboard / Page 6: Tooltip**
  This page enhances user interaction by providing additional details and contextual information when hovering over visuals, improving the overall dashboard experience.

Additionally, a Python-based AI chatbot was developed to allow users to interact with the data and obtain insights through simple queries.

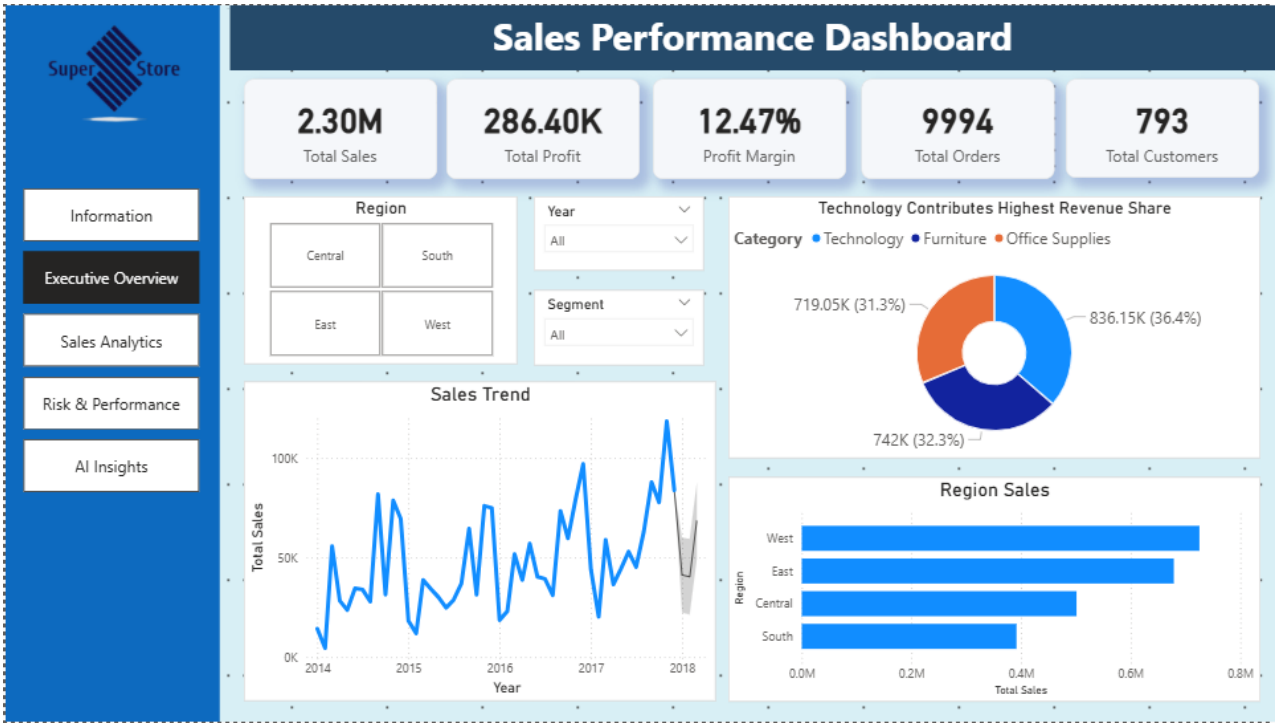


Figure 3: Power BI Dashboard – Executive Overview

Figure 4: Power BI Dashboard – Sales Analytics

4.3 AI Tool Integration Results

The project utilized an AI-based chatbot developed using Python to enhance data interaction and provide quick insights.

- **Tool used:**
  Python-based AI Chatbot
- **Prompts used:**
  1. Show total sales and profit from the dataset.
  2. Which region has the highest sales performance?
  3. Identify top-performing products based on sales.
- **Output quality:**
  The chatbot provided accurate and relevant responses based on the dataset. The outputs were clear, easy to understand, and useful for quickly retrieving key insights without manually analyzing the data.
- **How it enhanced the project:**
  The integration of the AI chatbot improved the overall efficiency of the project by enabling quick access to insights through simple queries. It reduced the need for manual analysis and made the system more user-friendly, especially for non-technical users

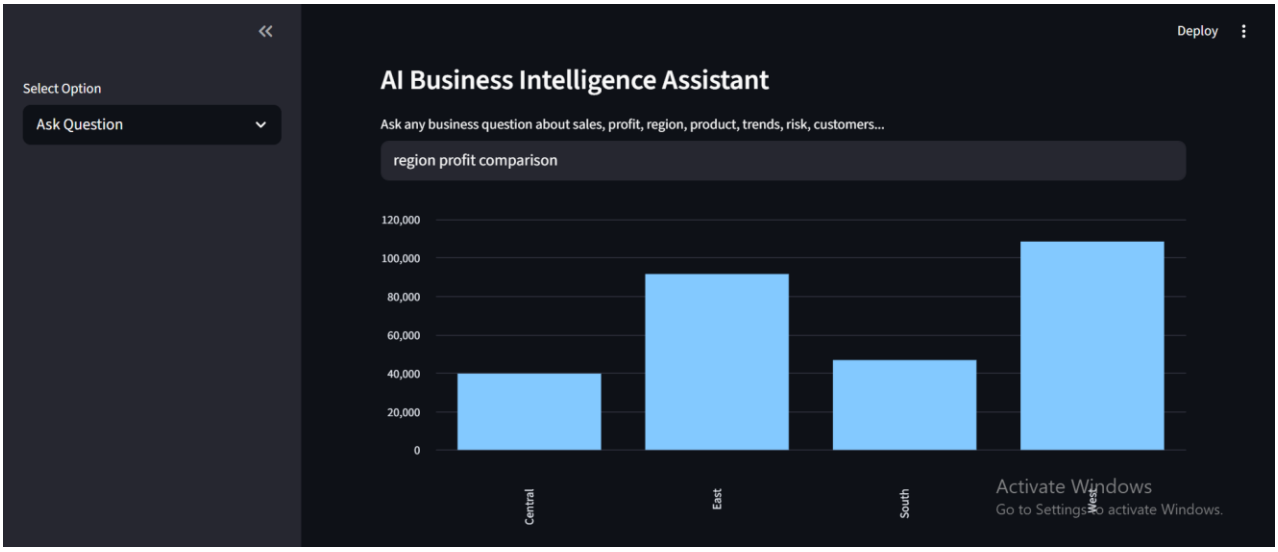


Figure 5: Screenshot of AI chatbot providing insights based on user queries

CHAPTER 5

Discussion and Conclusion

5.1 Key Findings

- After data cleaning, duplicate records and inconsistencies were removed, which improved the accuracy and reliability of the dataset for analysis.
- The Power BI dashboards clearly highlighted sales trends, making it easier to identify patterns and variations in sales performance.
- Regional analysis showed that certain regions performed better than others, helping to identify high-performing and low-performing areas.
- Product analysis revealed that a small number of products contributed significantly to overall sales and profit.
- The profit analysis helped in understanding the relationship between sales and profitability, identifying areas with high revenue but low profit.
- The integration of the Python-based AI chatbot enabled quick retrieval of insights, reducing the need for manual data analysis and improving efficiency.

5.2 Project Links

Share the relevant links to your project work for evaluation.

- GitHub Link of the Project: https://github.com/dhanalakshmi-97/AI-Powered-Sales-Business-Intelligence-Dashboard
- Video Recording of Project Demonstration:
- Power BI / Dashboard Link (if applicable): AI BI System Dashboard

5.3 Limitations

- The project is based on a sample retail dataset obtained from Kaggle and may not fully represent real-world business scenarios.
- The analysis is limited to historical data and does not include real-time data updates or live data integration.
- The project focuses only on descriptive analysis and does not include advanced predictive analytics or machine learning models.
- The AI chatbot developed is basic and may not handle complex queries or large-scale data efficiently.

- The Power BI dashboard is not published online and is limited to a local file, restricting accessibility.

5.4 Future Work

- Integrate real-time data updates using API connections to enable live tracking of sales data.
- Apply machine learning models such as sales forecasting and anomaly detection for advanced analysis.
- Improve the Python-based AI chatbot to handle more complex queries and provide more accurate insights.
- Develop a web-based interface to make the system more accessible for non-technical users.
- Expand the dataset by including more diverse and real-world data sources for better analysis.
- Enhance the Power BI dashboard with more advanced visuals and interactive features.

5.5 Conclusion

This project successfully demonstrated how a retail sales dataset can be cleaned, analyzed, and visualized using tools such as Microsoft Excel, Power BI, and Python. The main problem of unstructured and inconsistent data was addressed through effective data cleaning and preprocessing techniques.

The project achieved its objectives by transforming raw data into meaningful insights through interactive dashboards and integrating a basic AI chatbot for improved user interaction. The visualizations helped in identifying sales trends, regional performance, and top-performing products, supporting better decision-making.

The integration of the chatbot added value by enabling quick and easy access to insights, making the system more user-friendly. All the project objectives were successfully achieved.

Overall, this project enhanced my practical knowledge in data analytics, visualization, and basic AI implementation. It provided valuable hands-on experience and improved my understanding of how data-driven solutions can be applied in real-world business scenarios.

REFERENCES

[1] Microsoft Corporation. Microsoft Excel Documentation. Available at: https://support.microsoft.com/en-us/excel

[2] Microsoft Corporation. Power BI Documentation. Available at: Power BI Documentation

[3] Python Software Foundation. Python Documentation. Available at: https://docs.python.org/3/

[4] Edunet Foundation. AICW Fellowship Program Guidelines. 2026.

APPENDICES

Appendix A — Sample Dataset

1	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region	Product ID	Category	Sub-Category	Product Name	Sales	Quantity	Discount	Profit
2	1	CA-2016-152156	11-08-2016	11-11-2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42420	South	FUR-BO-10001798	Furniture	Bookcases	Bush Somerset Collection Bookcase	261.96	2	0	41.9136
3	2	CA-2016-152156	11-08-2016	11-11-2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42420	South	FUR-CH-10000454	Furniture	Chairs	Hon Deluxe Fabric Upholstered Stacking Chairs, Rounded Back	731.94	3	0	219.582
4	3	CA-2016-138688	06-12-2016	6/16/2016	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	California	90036	West	OFF-LA-10000240	Office Supplies	Labels	Self-Adhesive Address Labels for Typewriters by Universal	14.62	2	0	6.8714
5	4	US-2015-108966	10-11-2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311	South	FUR-TA-10000577	Furniture	Tables	Bretford CR4500 Series Slim Rectangular Table	957.5775	5	0.45	-383.031
6	5	US-2015-108966	10-11-2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311	South	OFF-ST-10000760	Office Supplies	Storage	Eldon Fold 'N Roll Cart System	22.368	2	0.2	2.5164
7	6	CA-2014-115812	06-09-2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	FUR-FU-10001487	Furniture	Furnishings	Eldon Expressions Wood and Plastic Desk Accessories, Cherry Wood	48.86	7	0	14.1694
8	7	CA-2014-115812	06-09-2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-AR-10002833	Office Supplies	Art	Newell 322	7.28	4	0	1.9656
9	8	CA-2014-115812	06-09-2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	TEC-PH-10002275	Technology	Phones	Mitel 5320 IP Phone VoIP phone	907.152	6	0.2	90.7152
10	9	CA-2014-115812	06-09-2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-BI-10003910	Office Supplies	Binders	DXL Angle-View Binders with Locking Rings by Samsill	18.504	3	0.2	5.7825
11	10	CA-2014-115812	06-09-2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-AP-10002892	Office Supplies	Appliances	Belkin F5C206VTEL 6 Outlet Surge	114.9	5	0	34.47
12	11	CA-2014-115812	06-09-2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	FUR-TA-10001539	Furniture	Tables	Chromcraft Rectangular Conference Tables	1706.184	9	0.2	85.3092
13	12	CA-2014-115812	06-09-2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	TEC-PH-10002033	Technology	Phones	Konftel 250 Conference phone - Charcoal black	911.424	4	0.2	68.3568
14	13	CA-2017-114412	4/15/2017	4/20/2017	Standard Class	AA-10480	Andrew Allen	Consumer	United States	Concord	North Carolina	28027	South	OFF-PA-10002365	Office Supplies	Paper	Xerox 1967	15.552	3	0.2	5.4432
15	14	CA-2016-161389	12-05-2016	12-10-2016	Standard Class	IM-15070	Irene Maddox	Consumer	United States	Seattle	Washington	98103	West	OFF-BI-10003656	Office Supplies	Binders	Fellowes PB200 Plastic Comb Binding Machine	407.976	3	0.2	132.5922
16	15	US-2015-118983	11/22/2015	11/26/2015	Standard Class	HP-14815	Harold Pawlan	Home Office	United States	Fort Worth	Texas	76106	Central	OFF-AP-10002311	Office Supplies	Appliances	Holmes Replacement Filter for HEPA Air Cleaner, Very Large Room, HEPA Filt	68.81	5	0.8	-123.858

Figure: Sample view of the raw retail sales dataset (first few rows).

Appendix B — Key Excel Formulas / DAX Measures Used

- **Total Sales**
  Total Sales = SUM (Sales_Fact [Sales])
  — Used to calculate total sales value

- **Total Profit**
  Total Profit = SUM (Sales_Fact [Profit])
  — Used to calculate overall profit

- **Total Orders**
  Total Orders = COUNT (Sales_Fact [Order ID])
  — Used to count total number of orders

- **Total Customers**
  Total Customers = DISTINCTCOUNT (Sales_Fact [Customer Name])
  — Used to count unique customers

- **Profit Margin**
  Profit Margin = DIVIDE ([Total Profit], [Total Sales])
  — Used to measure profitability

- **YoY Growth %**
  YoY Growth % = DIVIDE ([Total Sales] - [Sales LY], [Sales LY])
  — Used to calculate year-over-year growth

- **Top Product** — Identifies the product with highest sales using ranking logic

---

- **Product Rank**
  Product Rank = RANKX (ALL (Dim_Product [Product Name]), [Total Sales], , DESC)
  — Used to rank products based on sales

Appendix C — AI Prompts Used

- **Prompt 1:** "Show total sales, total profit, and total orders from the dataset."
- **Prompt 2:** "Which region has the highest and lowest sales performance?"
- **Prompt 3:** "Identify the top 10 products based on sales."
- **Prompt 4:** "Compare sales and profit to identify profitable and loss-making areas."
- **Prompt 5:** "Which category is the most and least profitable?"
- **Prompt 6:** "Show monthly sales trend and identify the best month."

Appendix D — Screenshot Gallery

Figure 1: Power BI Dashboard — Executive Overview

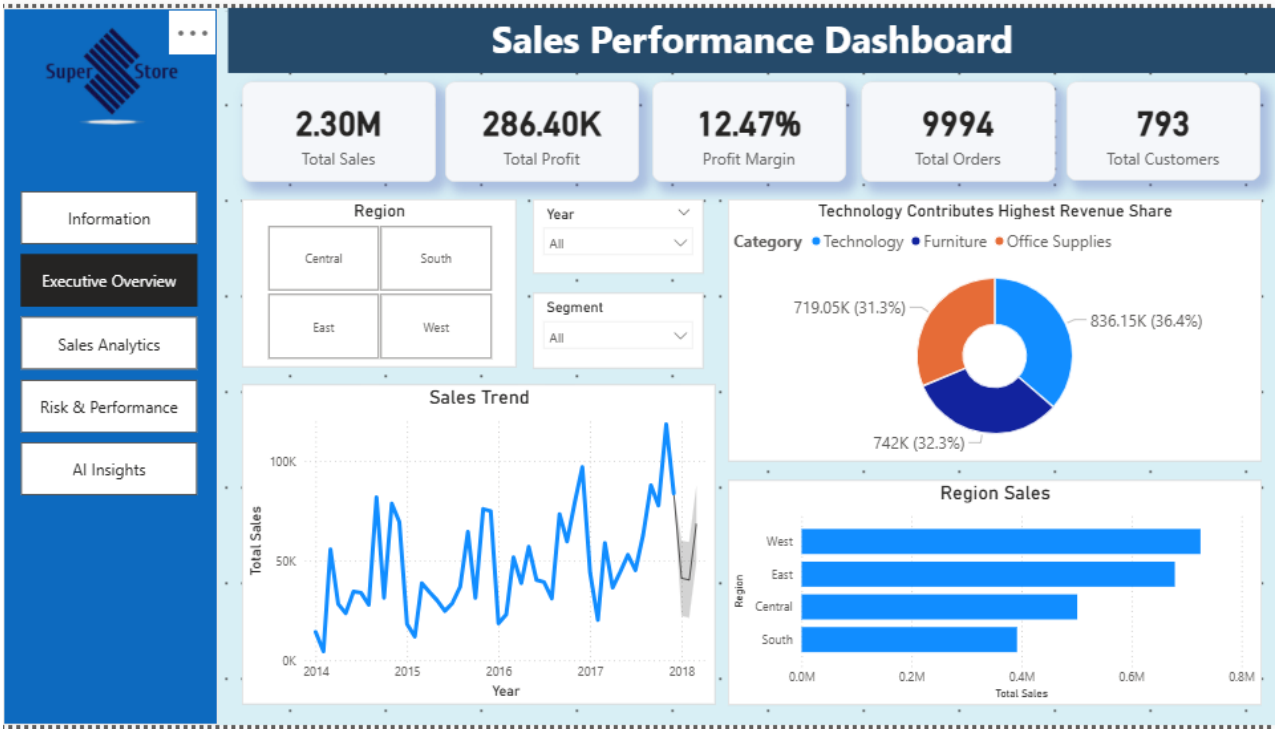
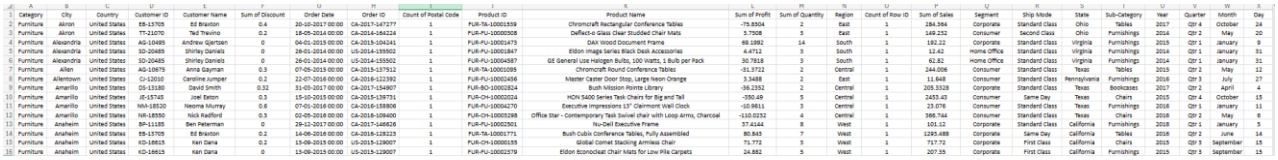


Figure 2: Power BI Dashboard — Sales Analytics

Figure 3: Power BI Dashboard — Risks and Performance

Figure 4: Cleaned Dataset in Microsoft Excel



	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
1	Category	City	Country	Customer ID	Customer Name	Sum of Discount	Order Date	Order ID	Count of Postal Code	Product ID	Product Name	Sum of Profit	Sum of Quantity	Region	Count of Row ID	Sum of Sales	Segment	Ship Mode	State	Sub-Category	Year	Quarter	Month	Day
2	Furniture	Akron	United States	EB-13705	Ed Braxton	0.4	20-10-2017 00:00	CA-2017-147277	1	FUR-TA-10001539	Chromcraft Rectangular Conference Tables	-75.8304	2	East	1	284.364	Corporate	Standard Class	Ohio	Tables	2017	Qtr 4	October	24
3	Furniture	Akron	United States	TT-21070	Ted Trevino	0.2	18-05-2014 00:00	CA-2014-164224	1	FUR-FU-10000508	Deflect-o Glass Clear Studded Chair Mats	5.7508	5	East	1	149.252	Consumer	Second Class	Ohio	Furnishings	2014	Qtr 2	May	20
4	Furniture	Alexandria	United States	AG-10485	Andrew Gjertsen	0	04-01-2015 00:00	CA-2015-104241	1	FUR-FU-10001473	DAX Wood Document Frame	68.1992	14	South	1	182.22	Corporate	Standard Class	Virginia	Furnishings	2015	Qtr 1	January	9
5	Furniture	Alexandria	United States	SD-20485	Shirley Daniels	0	26-01-2014 00:00	US-2014-135502	1	FUR-FU-10001847	Eldon Image Series Black Desk Accessories	4.4712	3	South	1	12.42	Home Office	Standard Class	Virginia	Furnishings	2014	Qtr 1	January	31
6	Furniture	Alexandria	United States	SD-20485	Shirley Daniels	0	26-01-2014 00:00	US-2014-155502	1	FUR-FU-10004087	GE General Use Halogen Bulbs, 100 Watts, 1 Bulb per Pack	30.7816	3	South	1	62.82	Home Office	Standard Class	Virginia	Furnishings	2014	Qtr 1	January	31
7	Furniture	Allen	United States	AG-10675	Anna Gayman	0.3	07-09-2015 00:00	CA-2015-137912	1	FUR-TA-10001595	Chromcraft Round Conference Tables	-31.3722	2	Central	1	244.008	Consumer	Standard Class	Texas	Tables	2015	Qtr 2	May	12
8	Furniture	Allentown	United States	CJ-12010	Caroline Jumper	0.2	22-07-2016 00:00	CA-2016-122392	1	FUR-FU-10002456	Master Caster Door Stop, Large Neon Orange	3.3488	2	East	1	11.648	Consumer	Standard Class	Pennsylvania	Furnishings	2016	Qtr 3	July	27
9	Furniture	Amarillo	United States	DS-13180	David Smith	0.32	31-03-2017 00:00	CA-2017-154807	1	FUR-BO-10002824	Bush Mission Pointe Library	-36.2352	2	Central	1	205.3328	Corporate	Standard Class	Texas	Bookcases	2017	Qtr 2	April	4
10	Furniture	Amarillo	United States	JE-15745	Joel Eaton	0.3	15-10-2015 00:00	CA-2015-139751	1	FUR-CH-10002024	HON 5400 Series Task Chairs for Big and Tall	-350.49	5	Central	1	2453.43	Consumer	Same Day	Texas	Chairs	2015	Qtr 4	October	15
11	Furniture	Amarillo	United States	NM-18520	Neoma Murray	0.6	07-01-2016 00:00	CA-2016-158806	1	FUR-FU-10004270	Executive Impressions 13" Clairmont Wall Clock	-10.9611	3	Central	1	23.076	Consumer	Standard Class	Texas	Furnishings	2016	Qtr 1	January	11
12	Furniture	Amarillo	United States	NR-18550	Nick Radford	0.3	02-05-2016 00:00	CA-2016-108400	1	FUR-CH-10002388	Office Star - Contemporary Task Swivel chair with Loop Arms, Charcoal	-110.0232	4	Central	1	366.744	Consumer	Standard Class	Texas	Chairs	2016	Qtr 2	May	8
13	Furniture	Anaheim	United States	BP-11185	Ben Peterman	0	29-12-2017 00:00	CA-2017-146826	1	FUR-FU-10002501	Nu-Dell Executive Frame	37.4144	8	West	1	101.12	Corporate	Standard Class	California	Furnishings	2018	Qtr 1	January	5
14	Furniture	Anaheim	United States	EB-13705	Ed Braxton	0.2	14-06-2016 00:00	CA-2016-128223	1	FUR-TA-10001771	Bush Cubix Conference Tables, Fully Assembled	80.845	7	West	1	1295.688	Corporate	Same Day	California	Tables	2016	Qtr 2	June	14
15	Furniture	Anaheim	United States	KD-16615	Ken Dana	0.2	15-09-2015 00:00	US-2015-128007	1	FUR-CH-10000155	Global Comet Stacking Armless Chair	71.772	3	West	1	717.72	Corporate	First Class	California	Chairs	2015	Qtr 3	September	15
16	Furniture	Anaheim	United States	KD-16615	Ken Dana	0	13-09-2015 00:00	US-2015-129007	1	FUR-FU-10002379	Eldon Econoclear Chair Mats for Low Pile Carpets	24.882	5	West	1	207.35	Corporate	First Class	California	Furnishings	2015	Qtr 3	September	15

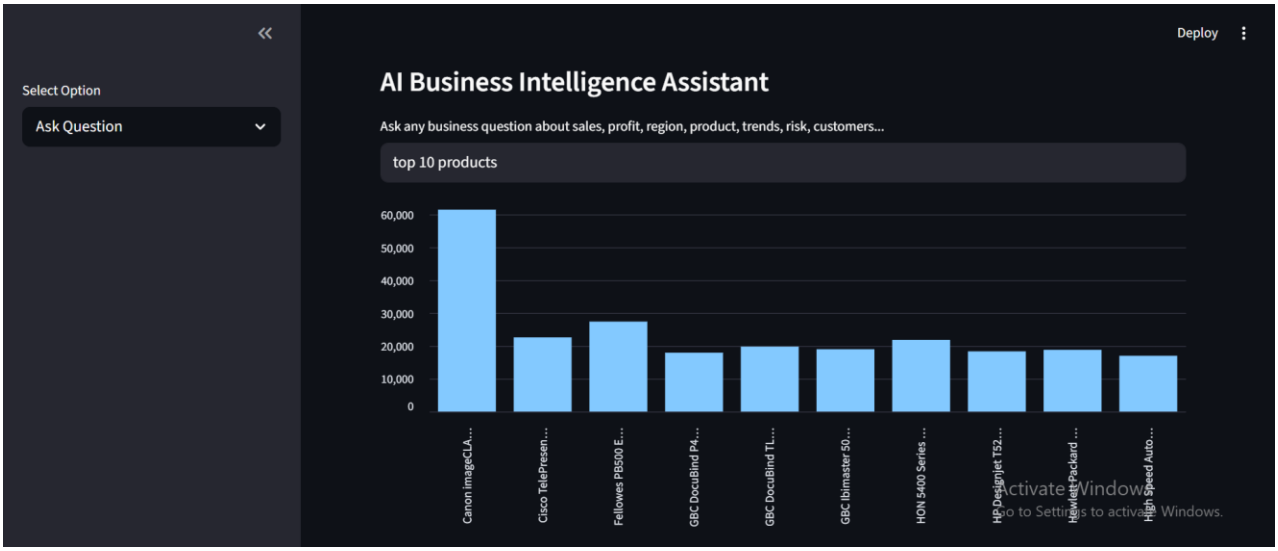



Figure 5: AI Chatbot Interaction